SpiderPhy: Multimodal Analysis of Arachnophobia

Physiological Data Processing Pipeline and Mixed-Effects Modeling

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2026-04-19

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Abstract

This project implements an end-to-end multimodal analysis pipeline based on the SpiderPhy open-access database (Lor et al.), a psychophysiological dataset collected from $N = 54$ spider-fearful individuals exposed to 174 phobic stimuli. Raw LSL-recorded signals were extracted from .mat files and processed in Python using scipy and NeuroKit2, yielding eight variables: heart rate, RMSSD, cardiac deceleration, SCR amplitude, respiratory variability, pupil diameter, pupil dilation speed, and gaze dispersion. Gaze data were corrected for tracker drift and scaling prior to heatmap generation. Variables were then modelled in R using linear mixed-effects models with parsimonious random structure selection ($\alpha = .05$). Within-person fear ratings robustly predicted heart rate, cardiac deceleration, respiratory variability, and pupil dilation speed. Phobia severity moderated the cardiac autonomic response to feared stimuli: more phobic individuals showed paradoxically higher heart rate variability (RMSSD) on their highest-rated trials, consistent with active parasympathetic regulation rather than passive hyperarousal. This result survived Benjamini-Hochberg correction across all four psychometric scales examined.

1 Introduction

This project implements a comprehensive end-to-end analysis pipeline based on the **Spider-Phy** open-access database (Lor et al.). The primary objective is to explore how psychological distress manifests across various biological channels during exposure to phobic stimuli (spiders) & to study the evolution of those data points as repeated exposure to phobic stimuli occurs.

1.1 The Dataset

The study involves $N = 54$ arachnophobic participants. During the experiment, they were exposed to: 174 spider-related images and 16 neutral stimuli.

Physiological Measures: ECG (Heart Rate), EDA (Skin Conductance), Respiration, and Oculometry (Gaze Tracking and Pupillometry).

Psychometric Measures: Standardized scales including FSQ, SAS, SPQ, and STAI.

1.2 Project Goals

1. **Pipeline Automation**: Building a workflow in Python and R to clean and process the experiment's data.
2. **Statistical Modeling**: Applying Linear Mixed-Effects Models (LMM) to account for both inter-individual and intra-individual variability.
3. **Data Visualization**: Generating spatial density overlays (heatmaps) to represent gaze distribution of the complete sample.

2 Data Preprocessing Pipeline

Raw physiological signals were gathered from 54 .mat files, each associated to a participant. Those files contained 3 streams of data: "pupil_capture": gaze coordinates, measurement confidence; "BrainVision RDA": ECG, GSR, Pulse, Respiration; "fear_stream": event codes, time stamp associated. Each of those stream, of course, were carrying a synchronized column indicating time in milliseconds.

2.1 Physiological Signal Cleaning (ECG & GSR)

Raw biosignals are inherently noisy (baseline wander, powerline interference, motion artifacts). To ensure data quality, I implemented a preprocessing workflow in Python using the `scipy.signal` and `NeuroKit2` libraries.

2.1.1 ECG

The primary cleaning step involved a **Butterworth filter**. I used a second-order, zero-phase (forward-backward) filter to avoid phase distortion, crucial for maintaining the temporal precision of cardiac R-peaks.

```
#| eval: false
#| code_summary: "ECG filtering logic"

def apply_butterworth_filter(data, low, high, sampling_rate, order):
    nyquist = 0.5 * sampling_rate
    sos = butter(order, [low / nyquist, high / nyquist], btype="band", output="sos")
    return sosfiltfilt(sos, data)
```

The Nyquist frequency is the maximum frequency that can be represented by a sampling rate without loosing fidelity. Higher isn't always better, as compute time can increase exponentially for each bits of information added.

2.1.2 GSR

The Galvanic Skin Response is provided. To exploit this data, several operations must be performed. It must be standardized due to inter-participant variability. Once done, another step must be undergone. GSR can be decomposed into two factors: the Tonic component & the Phasic component.

$$SC(t) = T(t) + P(t)$$

The first one is related to the typical behavior of a drop of sweat at the surface of the skin. The latter is related to the changes in GSR. The following filter must be applied:

$$IRF(t) = (e^{-\frac{t}{\tau_1}} - e^{-\frac{t}{\tau_2}})$$

The function `eda_phasic` from the package `neurokit2` must be used. It performs a Fourier transform to switch from temporal to frequencies referentials (allows the use of multiplication instead of convolution, which is way faster).

$$X_k = \sum_{n=0}^{N-1} x_n e^{-i2\pi \frac{kn}{N}}$$

Once this transformation done, a reverse Fourier transform is applied to recover the signal in temporal referencial.

2.2 Spatial drift correction

Another challenge is to ensure that gaze data is corrected. Surface data wasn't available, meaning the instrument provides coordinates in a referential that is not function of the displayed images. I came to that realization while generating heatmaps for "fun". Here one example:

(a) Raw gaze data

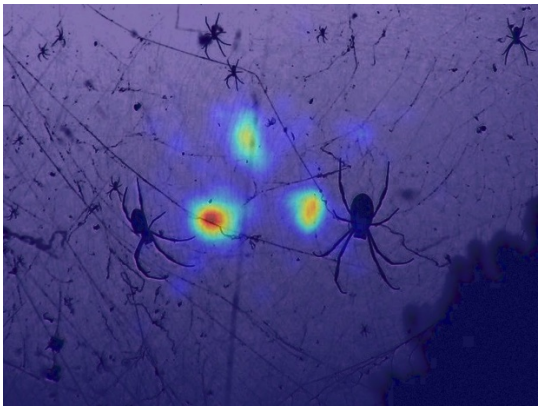
(a) Adjusted gaze data

What changed?

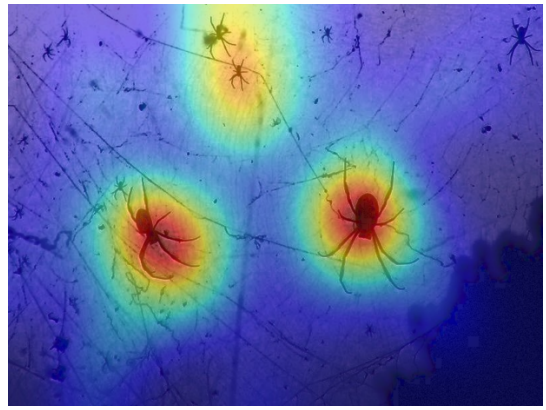
I decided to use the fixation cross event as a "ground truth" for each trials, taking into account only the last second of this event before the trial began.

```
#| eval: false
#| code-summary: "Gaze coordinates correction"

# 1. Extract gaze samples from the last 1s of the fixation period
mask_fix = (df_eye_raw["time"] >= (stim_start - 1.0)) & (df_eye_raw["time"] < stim_start)
fix_seg = df_eye_raw[mask_fix].copy()

# 2. Data Quality Check: Filter by confidence
fix_seg = fix_seg[fix_seg["confidence"] >= 0.6]

if not fix_seg.empty and len(fix_seg) >= 10:
    # 3. Compute drift offset relative to screen centre (0.5, 0.5)
    offset_x = fix_seg["x"].mean() - 0.5
    offset_y = fix_seg["y"].mean() - 0.5

    # 4. Apply correction to the subsequent stimulus window
    stim_seg["x"] = stim_seg["x"] - offset_x
    stim_seg["y"] = stim_seg["y"] - offset_y
```

Heatmaps were yet to be accurate. Stimuli containing multiple spiders helped me figure out what was the issue at hand.



(a) Incorrect Scaling



(a) Corrected Scaling (Scaling Factor = 4)

There is a scale issue. Using such stimuli as reference, I tested several scaling values until I found a proper fit.

```
#| eval: false
#| code-summary: "Scaling gaze coordinate referential"

SCALING_FACTOR = 4

# ...

# Expand gaze coordinates outward from the screen center (0.5, 0.5).
        # This corrects for the mismatch between the camera field of view and
        #   the actual display area, spreading points across the full image extent.
        points[:, 0] = (points[:, 0] - 0.5) * SCALING_FACTOR + 0.5
        points[:, 1] = (points[:, 1] - 0.5) * SCALING_FACTOR + 0.5
```

Problem solved! (note: time window is from 1.0 to 2.5 seconds into the trial to avoid center bias, also being forcefully removed in the script).

3 Physiological Feature Extraction

3.1 Heart Rate

The SpiderPhy project provided a reliable measure of heart rate through the use of a thoracic belt. Using the function `find_peaks` from `scipy.signal`, I was able to recover that dependent

variable for later analysis. (note: ECG data didn't allow for heart rate calculation due to incredible noise is several trials; manual cleaning would've been necessary).

3.2 RMSSD

The Root mean square of successive R-R differences, a barbaric term for a barbaric formula:

$$RMSSD = \sqrt{\frac{1}{N-1}\sum_{i=1}^{N-1}(RR_{i+1} - RR_i)^2}$$

Although threatening, here's how it works:

RR intervals are the time between two ECG peaks. They are squared to induce quadratic weighting as RR intervals increase. Each pairs of intervals (1 - n) are substracted (1 - 2, 2 - 3, ..., n-1 - n), squared, summed and divided by the number of pairs, then the root operation is applied to recover the millisecond metric.

A healthy heart is one with the ability to adapt to a changing environments. We should expect RMSSD to improve (increase) as exposure to spider stimuli continues.

3.3 Cardiac Deceleration

This variable is pretty straightforward, the trial is divided in two halves. For each halves the mean RR interval is computed, then subtracted. this provides us an overview of the changes in heart rate while first viewing the threatening stimuli and after we've processed it fully.

3.4 SCR Amplitude

As discussed previously, the GSR signal was pre-processed using the `neurokit2` package to only keep the Phasic component. With this data, we are able to compute the amplitude of the sweating, as well as the maximum sweating for a given trial.

```
#| eval: false
#| code-summary: "GSR Filtering"

gsr_signal = seg_physio["GSR_MR_100_xx"].values
        try:
            eda_decomposed = nk.eda_phasic(
                nk.standardize(gsr_signal), sampling_rate=SAMPLING_RATE
            )
            scr_phasic = eda_decomposed["EDA_Phasic"].values
```

```
            scr_amplitude = np.max(scr_phasic) - np.min(scr_phasic)
            scr_peak = np.max(scr_phasic)
```

We expect SCR amplitude to increase with fear rating: higher-rated stimuli should elicit stronger sympathetic activation, producing larger phasic responses. `scr_amplitude` (peak-to-peak range of the phasic signal) is retained as the dependent variable in subsequent models, as it captures the full magnitude of the electrodermal response per trial.

3.5 Pupil Dilation Speed

Rather than measuring absolute pupil diameter, this variable captures the change relative to the preceding baseline period. For each trial, the mean pupil diameter during the stimulus window is subtracted from the mean diameter recorded during the preceding fixation cross.

```
#| eval: false
#| code-summary: "Pupil dilation speed"

current_pupil = valid_eye[["diameter0_3d", "diameter1_3d"]].mean().mean()
res["pupil_dilation_speed"] = current_pupil - last_baseline_pupil
```

A positive value indicates dilation relative to baseline. We expect pupil dilation speed to increase with fear rating, reflecting greater arousal in response to more fear-inducing stimuli.

3.6 Gaze Dispersion

This variable quantifies how spatially scattered gaze points are across a trial. A focused gaze, concentrated on the spider, produces low dispersion, while an avoidant pattern produces high dispersion. It is computed as the root mean square distance of all valid gaze points from their centroid, using only samples recorded after the first second of stimulus onset to exclude the initial orienting response.

$$Dispersion = \sqrt{\frac{1}{n}\sum_{i=1}^{n} d_i^2}$$

```
#| eval: false
#| code-summary: "Gaze dispersion"

stable_eye = valid_eye[valid_eye["time"] >= (start_t + 1.0)]
centroid_x = stable_eye["norm_pos_x"].mean()
```

```
centroid_y = stable_eye["norm_pos_y"].mean()
distances = np.sqrt(
    (stable_eye["norm_pos_x"] - centroid_x) ** 2 +
    (stable_eye["norm_pos_y"] - centroid_y) ** 2
)
res["gaze_dispersion"] = np.sqrt(np.mean(distances ** 2))
```

The expected direction is less straightforward than for the other variables. Hypervigilance theories predict that phobic individuals fixate the threat more intensely (lower dispersion), whereas avoidance theories predict gaze redirection away from it (higher dispersion). Both patterns have empirical support in the literature, making this variable particularly interesting to examine.

4 Data Consolidation

4.1 Merging data sources

Once variables were extracted for each trial via the Python pipeline, three data sources were merged in R using `merge.R`:

- **Physiological features** (`processed_physio.csv`): one row per trial per participant, output of `process_mat.py`.
- **Stimulus ratings** (`rating_data/`): one CSV per participant, containing the fear rating given after each image, the `picture_id`, and trial metadata.
- **Psychometric scales** (`spiderPhy_beh_psy.xlsx`): one row per participant, containing pre- and post-exposure scores on FSQ, SAS, SPQ, and STAI.

```
#| eval: false
#| code-summary: "Three-source merge"

final_dataset <- all_stimuli |>
  left_join(physio_raw, by = c("participant_id", "trial_seq_id" = "trial_number")) |>
  left_join(psychometrics, by = "participant_id")
```

4.2 Cleaning process

Several cleaning steps were applied during this process. Baseline trials were removed, retaining only stimulus-onset rows. Only spider stimuli were retained for analysis (`condition_code == "Sp"`), excluding neutral images and control conditions.

4.3 Additional predictors

Two predictors were derived for use in mixed-effects models. The continuous fear rating (0-100) was decomposed into a within-person centered component (`rating_cwc`, capturing trial-by-trial fluctuations relative to each participant's own mean) and a between-person mean component (`rating_pm`, capturing stable individual differences in average fear level). Psychometric scales were grand-mean standardized to allow effect size comparison across instruments. The final dataset contains one row per spider trial per participant.

5 Psychometric Validation

Before entering the psychometric scales as predictors in the mixed-effects models, their measurement properties were examined in `psychometrics.R`. This step serves two purposes: verifying that the instruments behave as expected in this sample, and justifying which break-rating dimensions to retain as covariates.

5.1 Descriptive Statistics

The four composite scales show the expected profile for a spider-fearful sample. FSQ pre-exposure ($M = 54.4$, $SD = 19.4$) is consistent with moderate-to-severe fear, in line with the inclusion criterion ($FSQ \geq 24$). STAI-T ($M = 41.0$, $SD = 9.9$) reflects trait anxiety levels comparable to non-clinical norms, confirming that participants present spider-specific fear rather than generalised anxiety disorder.

```
#| eval: false
#| code-summary: "Descriptive statistics"

describe(composite_scales) |>
  select(n, mean, sd, median, min, max, skew, kurtosis)
```

5.2 Convergent Validity

FSQ, SAS, and SPQ all measure arachnophobia severity through different item formats; high inter-scale correlations are therefore expected. Pearson correlations confirmed this pattern, with all three spider-specific scales correlating strongly with one another. STAI-T showed moderate correlations with the phobia scales, consistent with the conceptual overlap between trait anxiety and phobic distress.

5.3 Internal Reliability of Break Ratings

Participants completed five emotional ratings after each of the four experimental blocks (Fear, Disgust, Excitement, Boredom, Exhaustion). Treating the four repeated measurements as indicators of a latent within-session construct, Cronbach's alpha was computed for each dimension.

Two dimensions were excluded a priori on theoretical grounds: Exhaustion and Boredom both show a directional upward trend across blocks, a non-stationarity that violates the assumption of a stable latent construct.

```
#| eval: false
#| code-summary: "Stationarity check"

psychometrics |>
  select(ID, starts_with("B")) |>
  pivot_longer(-ID,
    names_to  = c("run", "dimension"),
    names_pattern = "B(\\d)(.*)",
    values_to = "rating"
  ) |>
  ggplot(aes(x = run, y = M, color = dimension, group = dimension)) +
  geom_line() + geom_point() + geom_errorbar(...) +
  theme_minimal()
```

The remaining three dimensions: Fear, Disgust, and Excitement were retained following empirical confirmation of stationarity. Cronbach's alpha was acceptable for Fear and Disgust across the four runs. Excitement was retained despite theoretical ambiguity: visual inspection confirmed a flat trajectory across blocks, supporting its inclusion as a stable arousal indicator. These break ratings are available as covariates for future analyses.

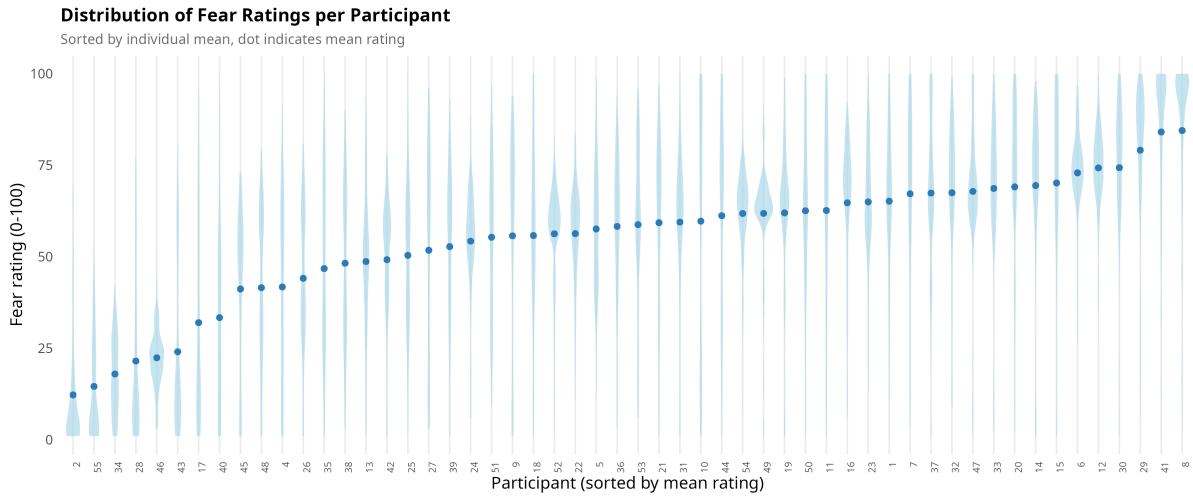


Figure 5: Distribution of fear ratings across participants, sorted by individual mean. Each violin represents the full distribution of ratings for one participant across all spider trials. The substantial inter-individual variability in mean rating level (ranging from ~10 to ~85) motivates the within-person centering applied in subsequent models.

6 Linear Mixed-Effects Models

Mixed-effects models were fitted in R using the `lme4` and `lmerTest` packages. The full modelling pipeline is implemented in `lmm.R`.

6.1 Model Specification

The data have a two-level nested structure: trials (Level 1) are nested within participants (Level 2), and each trial is associated with a specific stimulus image. A cross-classified random-effects structure was therefore adopted, with random intercepts for both participants and images:

$$Y_{ijk} = \beta_0 + \beta_1 \cdot \text{rating_cwc}_{ij} + \beta_2 \cdot \text{PSY}_j + u_{0j} + v_{0k} + \varepsilon_{ijk}$$

where i indexes trials, j participants, and k stimulus images. u_{0j} captures between-participant variance in baseline physiological response, and v_{0k} captures variance attributable to stimulus difficulty, some images are simply more fear-inducing than others, independently of who is viewing them.

The trial-level fear rating was decomposed into two components before modelling. The within-person centered component (`rating_cwc`) isolates trial-by-trial fluctuations relative to each

participant's own mean, capturing the momentary fear response. The between-person mean (`rating_pm`) captures stable individual differences in average fear level across the session. This decomposition prevents the conflation of within- and between-person effects, which would otherwise bias estimates in opposite directions. Both components were standardized prior to modelling. Psychometric scales were grand-mean standardized to allow effect size comparison across instruments with different native ranges.

6.2 Random Structure Selection

For each dependent variable, the random structure was selected parsimoniously via sequential likelihood-ratio tests ($\alpha = .05$), comparing three candidate models: random intercepts only, random intercepts with an uncorrelated random slope for `rating_cwc`, and random intercepts with a correlated random slope. The intercepts-only structure was retained for all eight dependent variables, indicating that the trial-by-trial effect of fear rating does not vary meaningfully across participants in this sample, at least not beyond what can be explained by the cross-level interactions examined below.

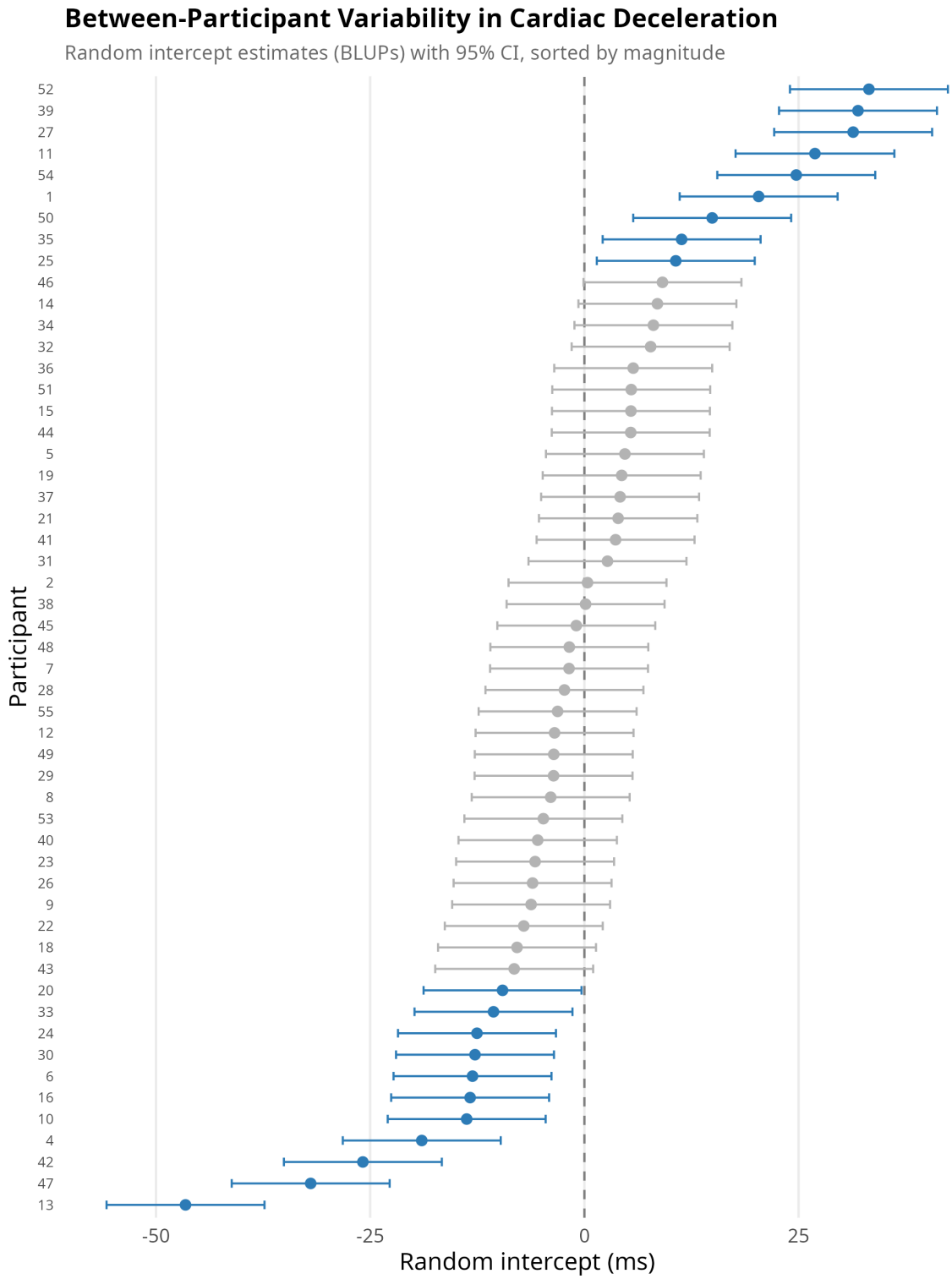


Figure 6: Between-participant variability in cardiac deceleration random intercepts (BLUPs), sorted by magnitude with 95% credible intervals. Blue points indicate participants whose interval does not overlap zero. The substantial spread confirms that baseline cardiac deceleration differs meaningfully across individuals, justifying the inclusion of random intercepts in all models.

```r
#| eval: false
#| code-summary: "Parsimonious random structure selection"

select_best_model <- function(m_simple, m_pente, m_full) {
  p_slope <- anova(m_simple, m_pente)[["Pr(>Chisq)"]][2]
  if (is.na(p_slope) || p_slope >= 0.05) {
    return(list(formula = formula(m_simple),
                type    = "Simple (intercepts only)"))
  }
  p_corr <- anova(m_pente, m_full)[["Pr(>Chisq)"]][2]
  if (!is.na(p_corr) && p_corr < 0.05) {
    list(formula = formula(m_full),  type = "Full (slope + corr)")
  } else {
    list(formula = formula(m_pente), type = "Slope (no corr)")
  }
}
```

Given the large number of models fitted (8 dependent variables $\times$ 4 psychometric scales $\times$ 2 model types = 64 models), p-values were adjusted using the Benjamini-Hochberg false discovery rate procedure to control the expected proportion of false positives among significant findings. Both uncorrected and FDR-adjusted p-values are reported.

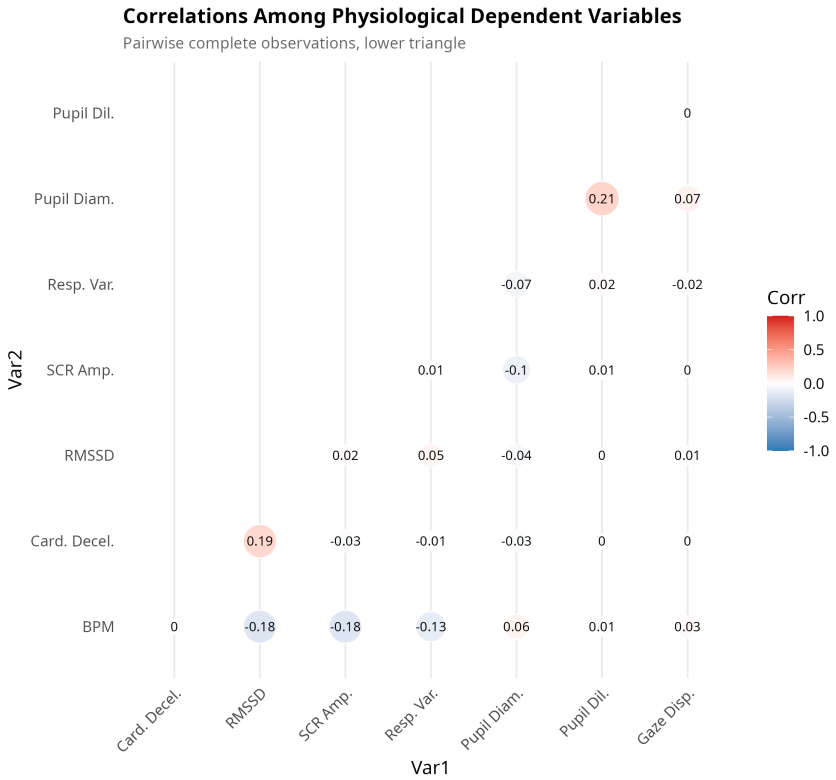


Figure 7: Pairwise correlations among the eight physiological dependent variables. The near-zero correlations across most pairs confirm that the DVs capture largely independent dimensions of the stress response, justifying their separate modelling rather than aggregation into a composite score.

6.3 Within-Person Effects of Fear Rating

Across all model specifications, the within-person centered fear rating (`rating_cwc_z`) was a robust predictor of four physiological variables. Higher-rated trials were associated with increased heart rate ($b = 0.45$, $SE = 0.08$, $p < .001$), stronger cardiac deceleration ($b = -4.24$, $SE = 0.69$, $p < .001$), greater respiratory variability ($b = 9.94$, $SE = 1.22$, $p < .001$), and faster pupil dilation ($b = 0.05$, $SE = 0.01$, $p < .001$). These effects survived FDR correction and are consistent with the physiological signatures of acute fear reported in the original SpiderPhy paper, providing a first validation that the custom extraction pipeline recovers meaningful signal.

RMSSD, SCR amplitude, raw pupil diameter, and gaze dispersion showed no reliable within-person effect of fear rating at the trial level.

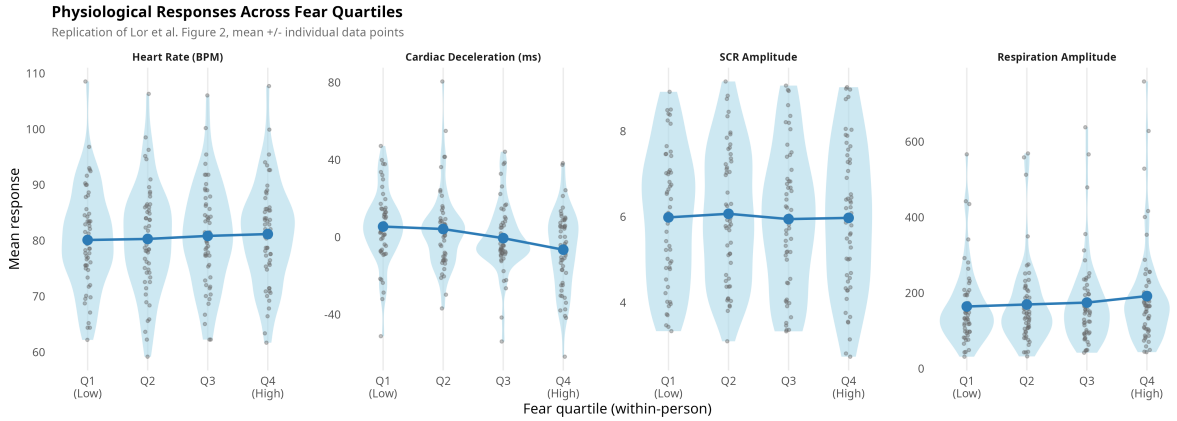


Figure 8: Replication of Lor et al: mean physiological response per within-person fear quartile (Q1 = lowest, Q4 = highest rated trials). Individual data points are shown in grey; blue points and lines indicate group means. Heart rate and respiration show the expected monotonic increase with fear level. Cardiac deceleration and SCR amplitude are flatter than in the original paper, consistent with the signal quality issues discussed in the limitations.

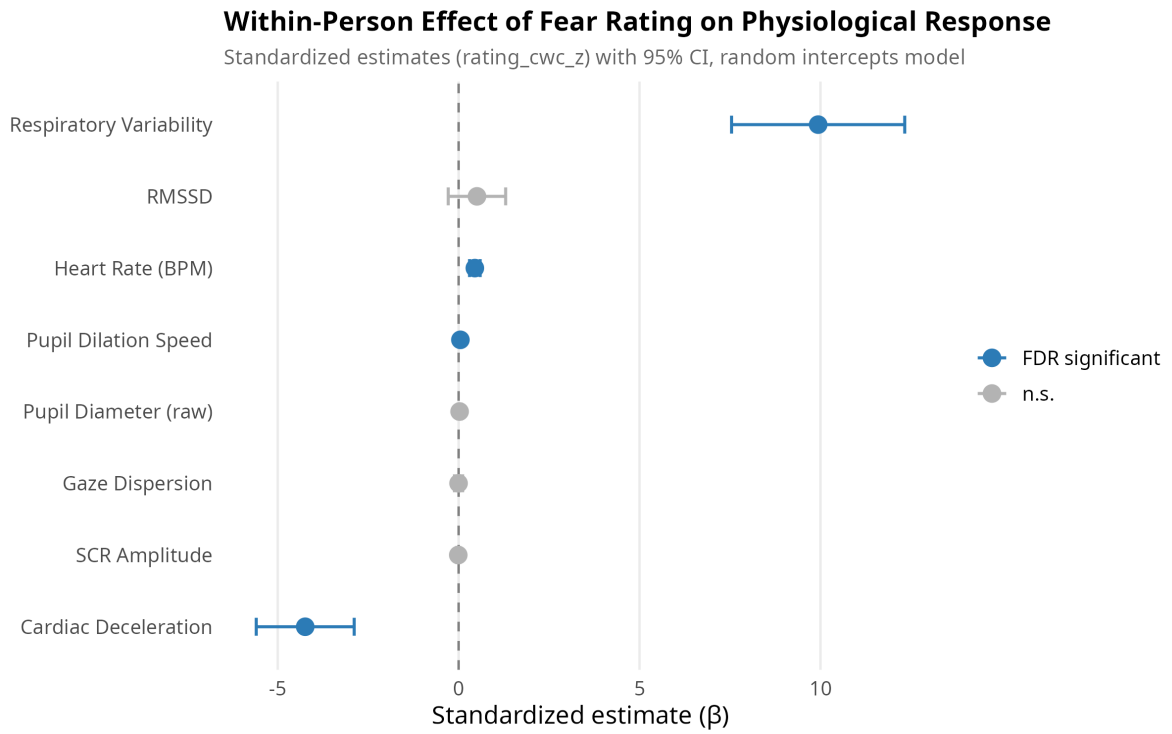


Figure 9: Standardized fixed-effect estimates (β) for the within-person fear rating predictor (`rating_cwc_z`) across eight physiological dependent variables, with 95% confidence intervals. Blue points indicate effects surviving FDR correction (Benjamini-Hochberg). Respiratory variability and cardiac deceleration show the largest effects; note that estimates are not directly comparable across DVs as dependent variables were not rescaled prior to modelling.

6.4 Between-Person Effects of Phobia Severity

No psychometric scale reliably predicted baseline physiological response levels after FDR correction. SPQ and SAS showed significant associations with raw pupil diameter and pupil dilation speed (ps = .026-.044), but these did not survive correction (p_FDR > .12). This null result suggests that phobia severity, as measured by questionnaire, does not straightforwardly elevate tonic physiological arousal, a finding consistent with research distinguishing trait fear from state reactivity.

6.5 Cross-Level Interactions

The most theoretically informative results concern the cross-level interactions between the within-person fear rating and between-person phobia severity. These test whether the trial-by-

trial physiological response to fear is amplified in more phobic individuals.

RMSSD. The interaction between `rating_cwc` and phobia severity was significant for all four psychometric scales, and all four survived FDR correction (p_FDR $< .001$ in each case). Contrary to a simple hyperarousal interpretation, more phobic individuals showed *greater* RMSSD on higher-rated trials, that is, *higher* heart rate variability in response to more frightening stimuli. This pattern is consistent with an active autonomic regulation process: rather than a blunted or overwhelmed cardiac response, highly phobic participants appear to recruit parasympathetic resources more strongly when facing their most feared stimuli. This result constitutes the primary original finding of this analysis.

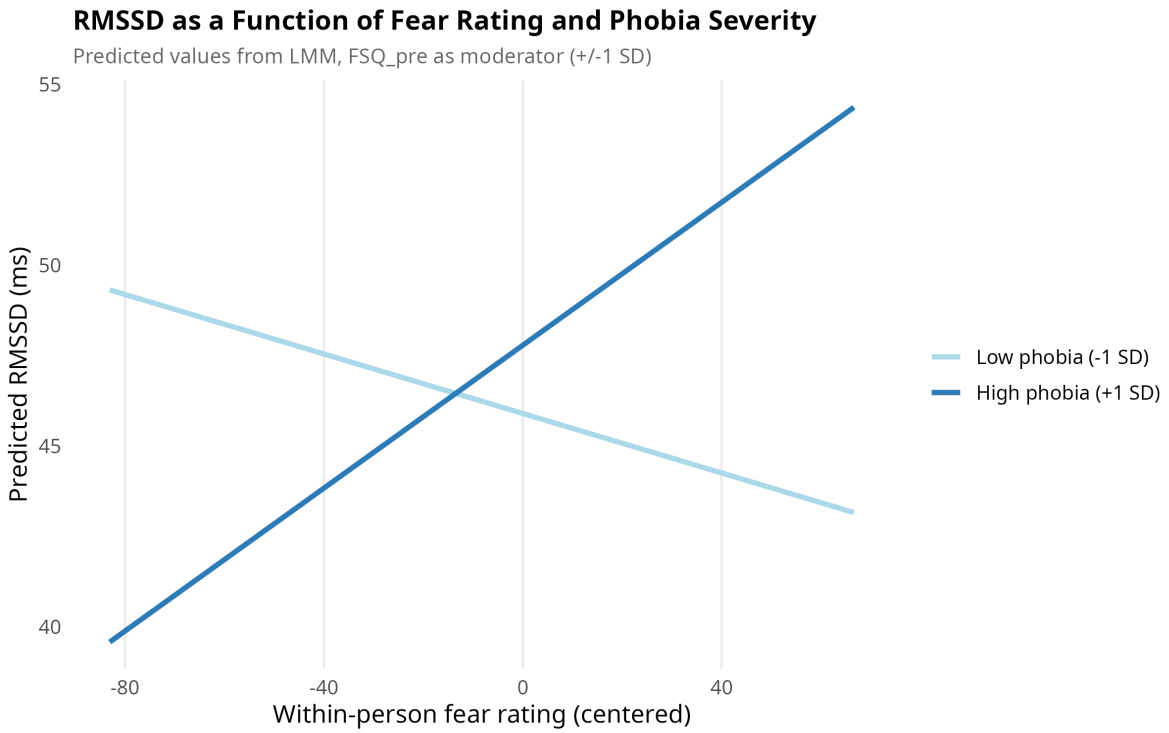


Figure 10: Predicted RMSSD as a function of within-person fear rating, separately for individuals low (-1 SD) and high ($+1$ SD) on FSQ pre-exposure score. The crossing pattern reflects the significant cross-level interaction: more phobic individuals show increasing cardiac variability on higher-rated trials, whereas less phobic individuals show the opposite trend.

Respiratory variability. The interaction between `rating_cwc` and SAS survived FDR correction ($b = 0.15$, $SE = 0.06$, p_FDR $= .036$), indicating that more phobic participants (as measured by SAS) show greater respiratory variability on high-fear trials. The same interaction with STAI was significant but did not survive correction.

SCR amplitude. Significant negative interactions were observed with SAS, SPQ, and STAI (ps = .013-.034), suggesting that more phobic or anxious individuals show smaller phasic skin conductance responses on higher-rated trials. However, none of these survived FDR correction and should be interpreted as exploratory tendencies only.

7 Limitations & Discussion

7.1 What this pipeline demonstrates

This project implements an end-to-end multimodal analysis pipeline, from raw LSL streams to mixed-effects models, on an open-access psychophysiology dataset. The within-person effects of fear rating on heart rate, cardiac deceleration, respiration, and pupil dilation replicate the physiological signatures reported in the original SpiderPhy paper.

The cross-level interaction on RMSSD is the most substantive original finding: phobia severity moderates the cardiac autonomic response to feared stimuli, with more phobic individuals showing paradoxically higher heart rate variability on their worst trials. This result aligns with theoretical accounts emphasising active coping over passive hyperarousal in arachnophobia.

7.2 Methodological limitations

SCR data quality. The original SpiderPhy authors excluded 29 of 54 participants from skin conductance analyses due to low signal quality and high noise levels. The present pipeline did not apply these exclusions, as the exclusion sheet was not retrieved from the OSF repository. SCR amplitude estimates for excluded participants are therefore likely to reflect noise rather than true electrodermal activity, which may explain the weak and inconsistent SCR effects observed.

Random structure. The parsimonious selection procedure retained random intercepts only for all dependent variables, meaning that individual differences in the slope of physiological response to fear were not modeled. This may reflect genuine homogeneity in within-person reactivity across participants, or it may reflect insufficient power to detect slope variance and after that auto-corellation.